

Problem-Solution Fit

Date	20 June 2026
Team ID	
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

1. CUSTOMER SEGMENT(S) [CS] * Regional disaster management cells and municipal authority boards. * Residents and property owners living in low-lying, flood-prone environments. * Local emergency first responders and safety organizations.	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] * Budget Constraints: Tiny local administrative boards possess small technology adoption budgets. * Device Dependencies: Ground-level emergency workers rely entirely on basic mobile devices over weak cell networks. * Data Barriers: Users lack data science capabilities to interpret complex weather files.	5. AVAILABLE SOLUTIONS PROS & CONS [AS] * General News Channels: Pros: Widespread reach. Cons: Too broad, lacking localized, analytical predictive depth. * Physical Ground Sensors: Pros: High immediate precision. Cons: Expensive deployment cost, prone to flood damage, zero proactive predictive sight. * Basic Weather Apps: Pros: Real-time information access. Cons: Do not map meteorological traits to explicit regional flood odds.
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] * Unexpected flash flooding leaves zero prep time for shifting assets. * Existing warning outlets offer wide regional general	9. PROBLEM ROOT / CAUSE [RC] * Traditional public setups track raw rainfall without calculating cascading environmental dynamics (e.g., cloud visibility,	7. BEHAVIOR + ITS INTENSITY [BE] * Behavior: Constantly tracking disparate government portals, checking seasonal water gauges manually, or relying on word-of-mouth chatter during storms.

summaries rather than localized danger risk profiles.

*** Frequency:** Highly recurring seasonal stress hitting communities multiple times during monsoons.

annual trends, seasonal saturation balances).

*** Advanced prediction models** remain siloed in expert tools rather than being linked to simple cloud-hosted interfaces.

*** Intensity: Extremely High.** Financial loss risks and physical isolation create high anxiety levels among targeted stakeholders.

3. TRIGGERS TO ACT [TR]

*** The onset of heavy seasonal monsoon storm cycles.**

*** Sudden spikes in local weather service rainfall alerts.**

*** Sudden visible flooding spikes occurring in neighboring administrative sectors.**

10. YOUR SOLUTION [SL]

*** A full-stack, cloud-hosted machine learning alert interface.**

*** Incorporates a pre-trained, high-accuracy XGBoost machine learning model that takes standard weather inputs and generates immediate "Flood Chance" indicators.**

*** Provides an accessible web form, rendering complex calculations as direct, clear guidance metrics.**

8. CHANNELS OF BEHAVIOR [CH]

ONLINE

*** Official mobile web dashboards, cloud application portals, and emergency communication threads.**

4. EMOTIONS BEFORE / AFTER [EM]

*** BEFORE:** Panicked, uncertain, exposed, and reactive during unpredictable climate situations.

*** AFTER:** Prepared, secure, informed, and proactive due to fast, verifiable, data-backed risk assessments.

OFFLINE

*** Community safety assemblies, megaphone emergency broadcasts, and physical flood barricade staging points.**